

LIFE-CRAFT: A Multi-Agentic Conversational RAG Framework for Lifestyle Medicine Coaching with Context Traceability and Case-Based Evidence Synthesis

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1 Introduction

Problem

Access to expert guidance in lifestyle medicine remains limited due to various socioeconomic and psychological barriers. Individuals often hesitate to seek in-person consultations due to stigma, cost, time constraints, or lack of immediate availability of trained professionals.

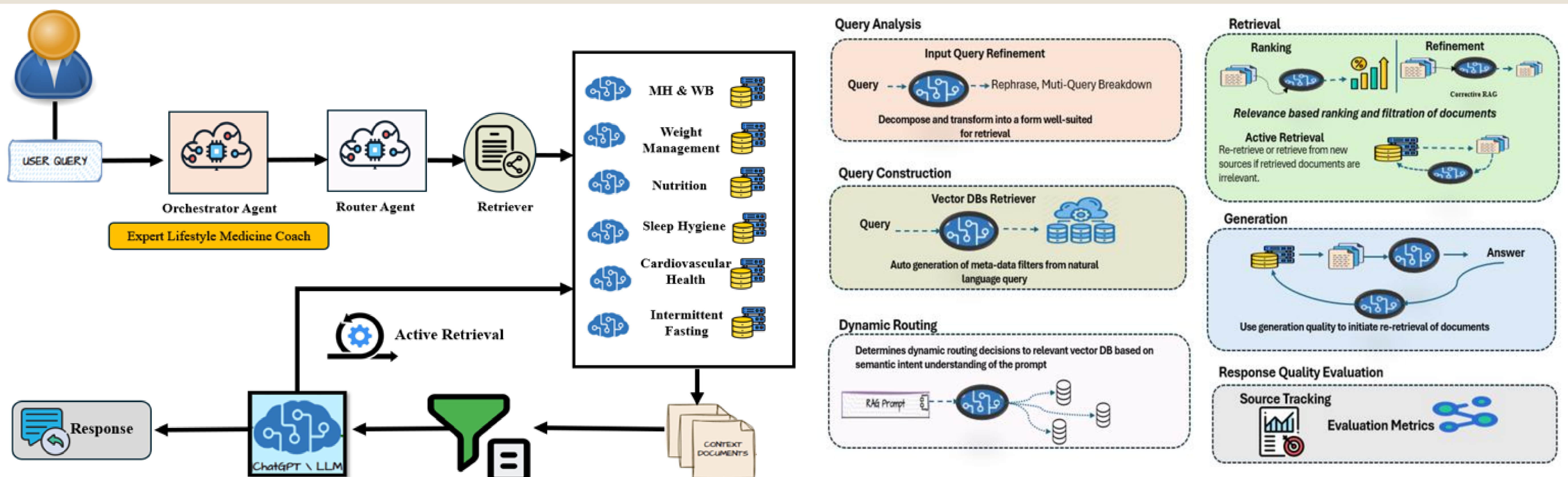
Aim

We present a novel multi-agentic RAG-framework specifically tailored for the lifestyle medicine domain. Our system acts as a virtual lifestyle coach, capable of delivering personalized, timely, and evidence-backed guidance grounded in behavioral science and lifestyle medicine literature. Our pipeline operates on a curated private repository of over 7,000 lifestyle medicine research PDFs, ensuring that responses are grounded in peer-reviewed, domain-specific evidence.

2 Key Highlights

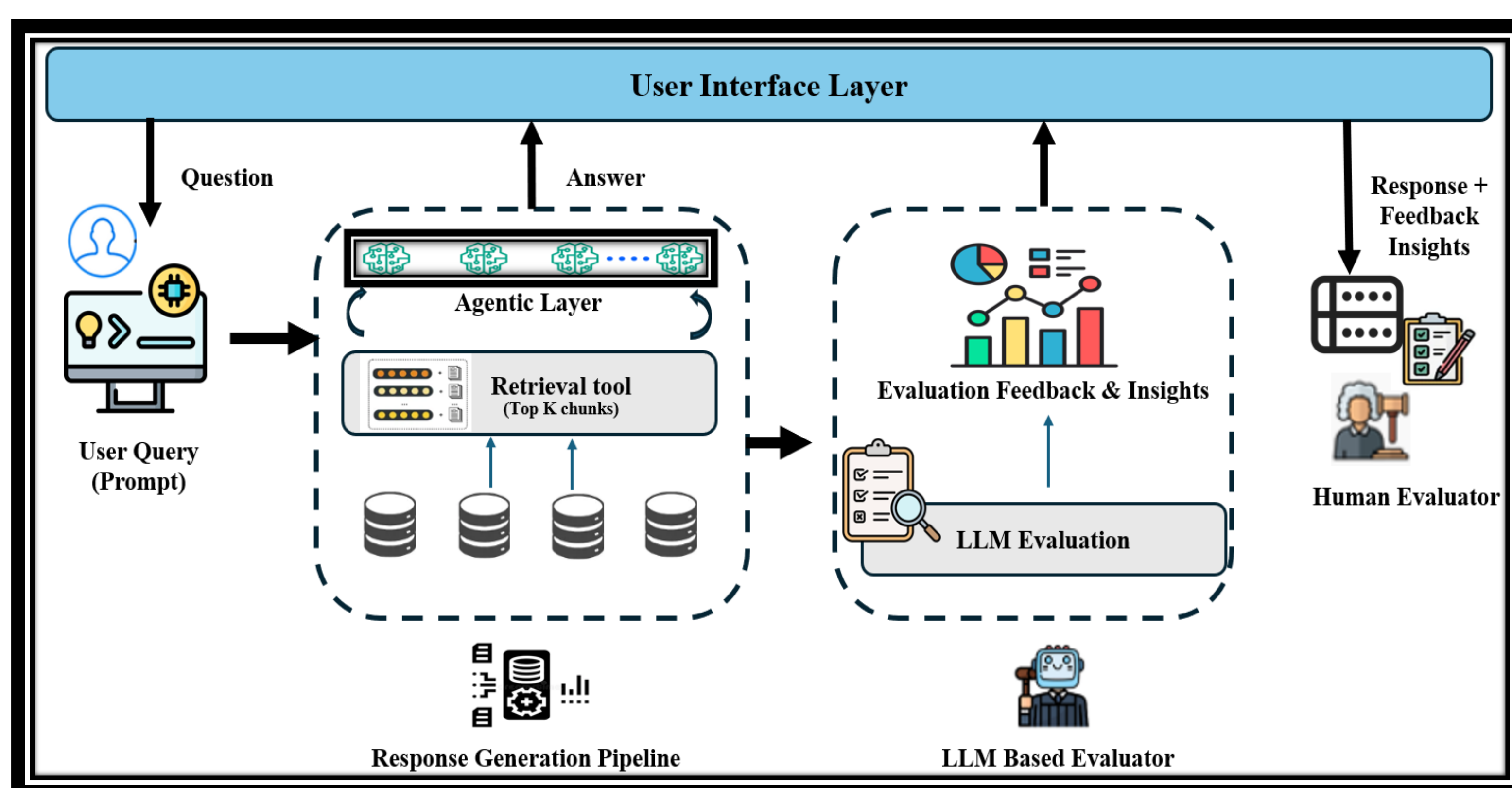
- **Domain-Aware Agentic Architecture:** Directs queries to the most relevant lifestyle medicine subdomain expert agents.
- **Reduced Hallucination and Enhanced Credibility:** Mitigates hallucinations, a common issue in general-purpose LLMs.
- **Traceability and Evidence Linking:** Each generated answer includes source tracing, enhancing transparency and user trust.
- **Iterative Refinement:** Multi-turn refinement for deeper coverage and tailored insights, unlike single-pass models.
- **Human and Automated Evaluation:** Dual-evaluation scheme with automated visual insights for LLM response verification.

3 Methods



- Lifestyle Medicine Expert Coach within multi-agent architecture** based on an orchestrator agent, hybrid routing algorithm.
- Corrective RAG Pipeline** is invoked to re-rank retrieved chunks based on confidence scores and source quality.
- Outputs undergo **LLM-based review** for **coherence, factual grounding, and source traceability**.

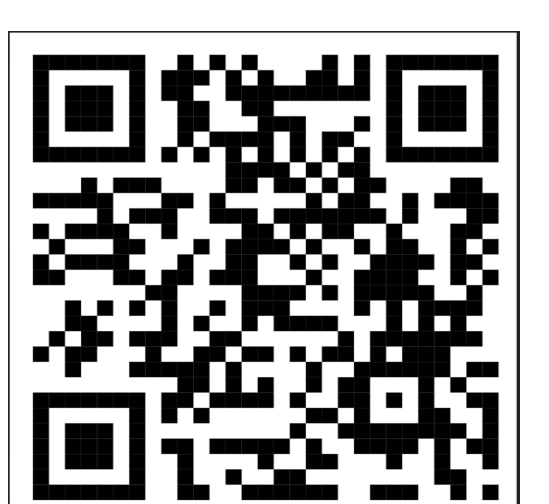
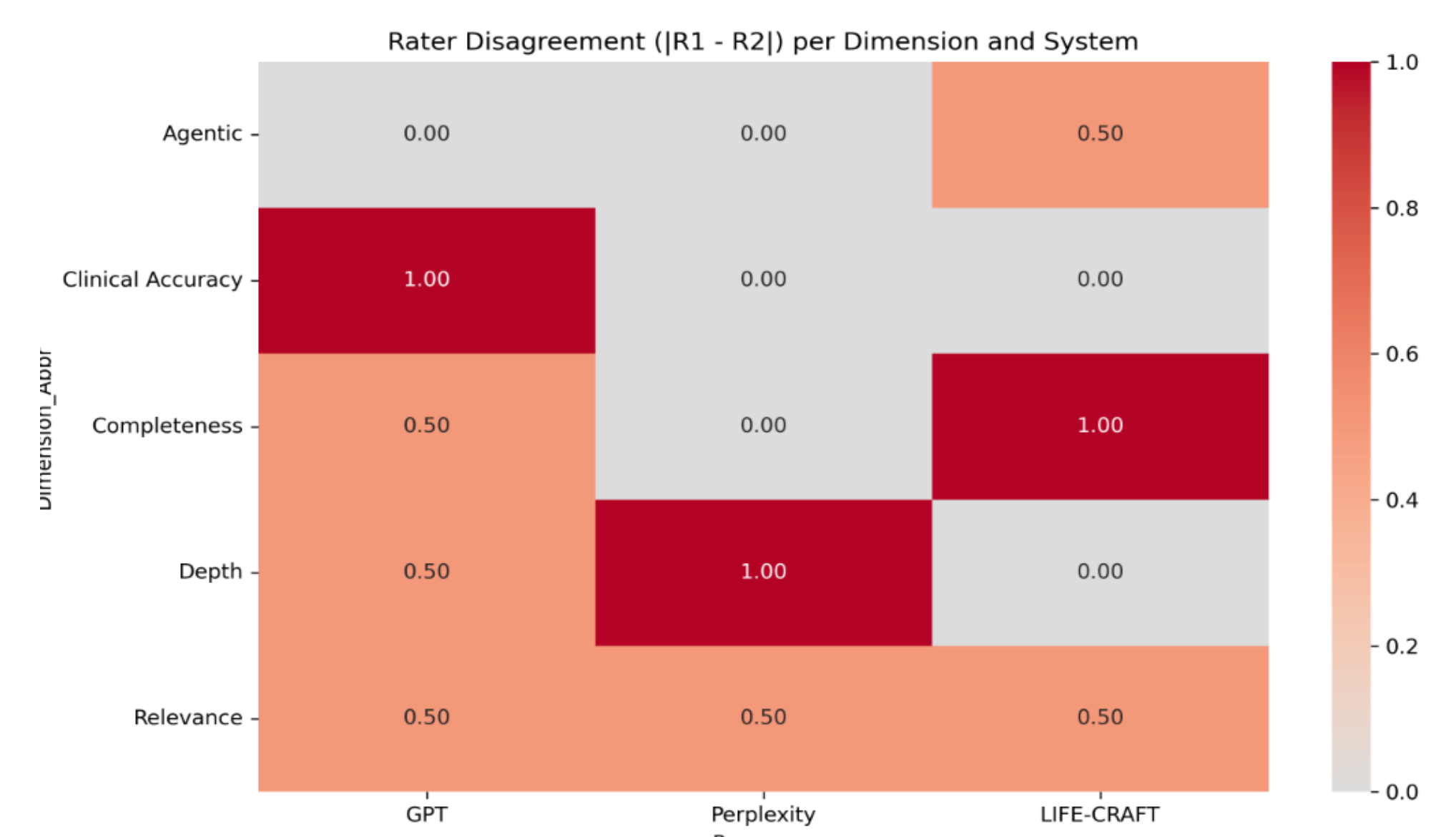
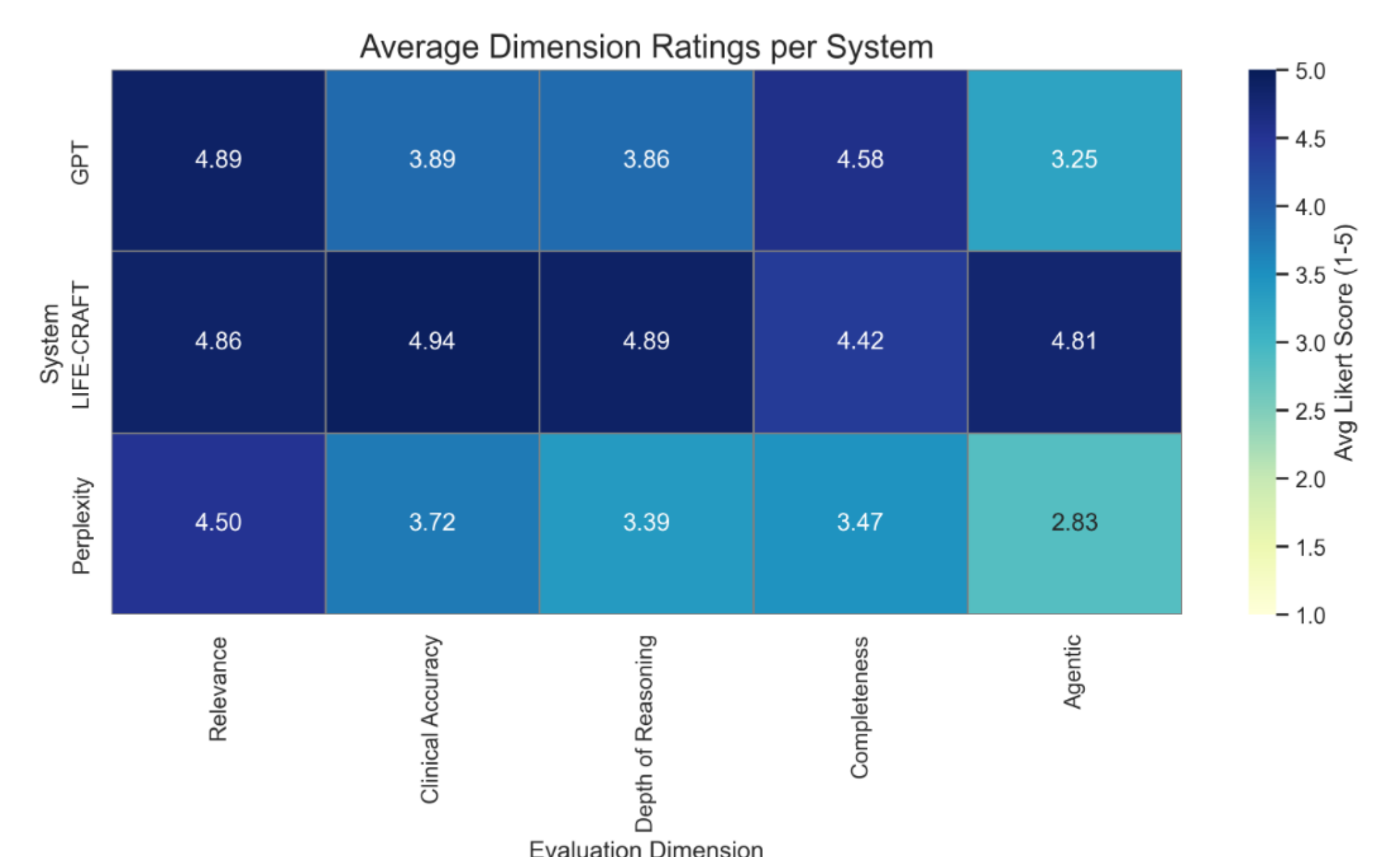
4 Results



- **Dual-evaluation strategy** for quality assessment of the generated responses using Cohen's kappa coefficient to **assess inter-rater agreement**

5 Conclusion

- **Novel framework** designed at the intersection of **powerful AI tools** and **lifestyle medicine domain** that may serve as a **virtual coach**.
- In future we intend to integrate **wearable devices data** and combine **passive sensing** with **domain-aware reasoning** for serving as an **indispensable tool** in **preventive healthcare** and **self-managed wellbeing**.
- **Comprehensive human centric behavioral ontology framework** underway to **characterize personas** and engineer the context of the LLMs for personalized responses.



Access paper